MIL-I-85080(AS)
28 November 1977

MILITARY SPECIFICATION

INSULATION SLEEVING, ELECTRICAL, NON HEAT SHRINKABLE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for chemically expanded, flexible, electrical insulating sleeving that shrinks to a predetermined size upon solvent evaporation.

1.2 Classification. The shrinkable sleeving shall be as specified in the applicable military specification sheet.

1.2.1 Colors. Colors, when applicable, shall be identified in accordance with the following:

Designator	Color	Designator	Color
0	Black	6	Blue
1	Brown	7	Violet (Purple)
2	Red	8	Gray (Slate)
3	Orange	9	White
4	Yellow	C	Clear
5	Green		

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5970

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1.2.2 Part number designation. Part number designations under this specification shall be as follows:

M85080/XX	-XXX	-X
Applicable Specification Sheet	Class (See Applicable Specification Sheet) Size Identifier	Color Designator (sse 1.2.2)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

SS-S-550	Sodium Chloride, Technical, For Water Softening Units
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corners
PPP-B-636	Box, Shipping, Fiberboard
PPP-C-96	Can, Metal, 28 Gauge and Lighter

MILITARY

MIL-B-117	Bag, Sleeve and Tubing, Interior Packaging
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8243	Anti-icing and Deicing-Defrosting Fluid
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-C-43616	Cleaning compound, Aircraft Surface

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STANDARDS

FEDERAL

FED-TEST-MTHD-STD-228	Cable and Wire, Insulated, Methods of Testing

MILITARY

MIL-STD-104	Limit for Electrical Insulation Color
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-120	Gage Inspection
MIL-STD-129	Marking for Shipment and Storage

SUPPLEMENT

MIL-I-85080/1(AS)	Insulation Sleeving, Electrical, Nonheat Shrink, Polyvinyl Chloride, Flexible Non-Crosslinked
MIL-I-85080/2(AS)	Insulation Sleeving, Electrical, Nonheat Shrink, Silicone, Rubber, Flexible

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents from a part of this specification to the extent specified herein, Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 149-75	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.
D 374-74	Tests for thickness of Solid Electrical Insulation
D 412-75	Tension Testing of Vulcanized Rubber
D 570-63	Testing for Water Absorption of Plastics
D 638-72	Test for Tensile Properties of Plastic

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D 876-71 Testing for Nonrigid Vinyl Chloride Polymer Tubing

D 882-73 Test for Tensile Properties of thin Plastic Sheeting

G 21-70 Recommended Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Data. No data is required by this specification or by the applicable documents referenced in section 2 of this specification unless specified by the contracting officer.

3.3 Toxicity. The material shall have no adverse affect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as advisor to the procuring agency.

3.4 Dimensions. Dimensions and tolerances for "as supplied" ID, after unrestricted shrinkage ID, wall thickness and longitudinal change shall be measured as specified in 4.6.4 and be in accordance with the applicable specification sheet.

3.5 Unit of Issue. Procurement documents should specify unit of issue as foot. Packaging "put-ups" shall be in lengths as specified in the contract or order (See 6.2).

3.6 Properties. Properties shall be as specified in Table I. Unless otherwise specified, data shall be obtained on recovered (shrunk) sleeving.

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Table I. Properties

Property	Requirement	Test Method
Water absorption	Specification sheet	4.6.2 ASTM D 570 2/
Dielectric Strength	Specification sheet	4.6.2 ASTM D 876
Volume Resistivity	Specification sheet	4.6.2 ASTM D 876
Fungus resistance	No growth	4.6.2 ASTM G 21
Color	MIL-STD-104	Visual inspection
Voltage withstand	Specification sheet	4.6.5
Low Temperature flexibility	Specification sheet	4.6.6
Heat shock	Specification sheet	4.6.7
Heat resistance	Specification sheet	4.6.8
Corrosion	Specification sheet	4.6.9 1/
Fluid resistance	Specification sheet	4.6.10
Odor	No objectionable pungent aroma	4.6.11
Modulus	Specification sheet	4.6.12
Tensile strength	Specification sheet	4.6.13
Ultimate elongation	Specification sheet	4.6.13
Flammability	Specification sheet	4.6.14
Color stability	Conforms to as received limits	4.6.15

1/ See specification sheet for test method

2/ When testing small (ID) sleeving, special care shall be taken to remove all surface water from the inner wall before reweighing.

3.7 Shelf life The manufacturer shall certify that the sleeving in original unopened containers, when stored as specified in 4.6.16 shall conform to the dimensional requirements of 3.4

3.8 Workmanship. When examined visually, the sleeving shall be free from voids, blisters, lumps, tears, pinholes, seams, foreign matter and other defects which could affect performance before, during and after shrinkage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The examination and testing of the sleeving shall be classified as Quality Conformance inspection (See 4.3).

4.3 Quality conformance

4.3.1 Lot formation. Unless otherwise specified, a lot shall consist of all the non-heat shrink sleeving of one specification sheet, class, color and size that has been processed into the sleeving from one batch and submitted for inspection at one time.

4.3.2 Sampling

4.3.2.1 Visual and dimensional inspection of end item. Samples for the inspection shall be selected in accordance with Inspection Level S-3 of MIL-STD-105. The samples shall be selected just prior to cutting for "packaging put-up". The sampling unit shall be one foot of sleeving. Each sample shall be inspected as specified in 4.3.3.1.

4.3.2.2 Physical property tests. The lot size for this inspection shall be the total amount of sleeving of one specification sheet and class offered for inspection. Thirty feet of sleeving shall be selected at random from each lot and tested as specified in 4.3.3.2. Sleeving selected and visually inspected in 4.3.3.1 may be used to "make-up" the sample for tests.

4.3.2.3 Preparation for delivery. A quantity of shipping containers fully prepared for delivery, just prior to closure, shall be selected at random from each lot in accordance with Inspection Level S-2 of MIL-STD-105. The lot size shall be the number of shipping containers.

4.3.3 Inspections and tests.

4.3.3.1 Visual and Dimensional inspections. Samples selected in 4.3.2.1 shall be examined for conformance to the requirements in Table II. The Acceptable Quality Level (AQL) for this inspection shall be as specified in Table II. The number of determinations on each sample shall be as specified in Table IV.

TABLE II VISUAL AND DIMENSIONAL INSPECTION

Inspection	Requirement	AQL
Workmanship	3.8	2.5 As supplied ID
After unrestricted shrinkage	3.4	1.0
ID	3.4	1.0
Wall thickness	3.4	1.0
Longitudinal/change	3.4	1.0
Color	Table I	1.0

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4.3.3.2 Physical property tests. Test specimens shall be prepared from the sample selected in 4.3.2.2 and tested to the requirements of Table I. Whenever possible, the material utilized for 4.3.3.1 shall be used as test specimens for this inspection. The number of specimens tested and reporting of results shall be in accordance with Table IV. Nonconformance of a specimen to any requirement shall be cause to reject the lot represented by the sample.

4.3.3.3 Packaging inspection. Each sample selected in accordance with 4.3.2.3 shall be examined for the defects specified in Table III and to all other requirements of section 5 of this specification. The AQL for this inspection shall be 4.0. In addition, shipping containers fully prepared for delivery shall be examined after closing for closure defects.

Table III. Packaging Inspection

Examine	Defect
Packaging	Not level required by contract or purchase order
	Material or construction not as specified.
Packing	Not level required by contract or purchase order.
	Material or construction not as specified.
	Any nonconforming component, incomplete closures, inadequate strapping, bulged or damaged containers.
Count	Less than specified or indicated quantity per shipping container.
Markings	Warning labels or marking missing.
	Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.4 Manufacturer's Data. For each procurement the manufacturer shall include with data required by the contract or order, a certification that the sleeving meets or exceeds the shelf life requirement of 3.7 and 4.6.16.

4.5 Report of test results. Table IV shall be used to prepare test reports. The number of specimens per test and method of reporting the results are contained therein.

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4.6 Test methods.

4.6.1 Test conditions

4.6.1.1 General. Unless otherwise specified herein, the sleeving, in packages, and measurement gages shall be conditioned for 4 hours at 23 ± 1°C (73.4 ± 2°F) and a relative humidity of 50 ± 2 percent prior to testing. When the sleeving is tested after shrinkage conditioning of the sleeving shall take place after completion of the shrinking procedure. Unless otherwise specified in the specification sheet, an oven with a circulating air velocity of 100 to 200 feet per minute shall be used whenever heating is required.

Table IV. Quality conformance inspection

Property	Requirements	Number of Determinations Per Sample	Results Reported as	
			Pass or Fail 1/	Average or Multiple Determinations 2/
Inside diameter, as supplied	3.4	Two	X	
Inside diameter, after unrestricted shrinkage	3.4	Two	X	
Wall thickness, after unrestricted shrinkage.	3.4	Two	X	
Longitudinal change	3.4	Two	X	
Color	3.6	One	X	
Color Stability	3.6	Three	X	
Workmanship	3.8	One	X	
Fungus resistance	3.6	Three	X	
Corrosion	3.6	Three	X	
Odor	3.6	Three	X	
Tensile Strength	3.6	Three		X
Ultimate elongation	3.6	Three		X
Dielectric strength	3.6	Three	X	
Volume resistivity	3.6	Three	X	
Low temperature flex	3.6	Three	X	
Heat resistance	3.6	3/		3/
Water absorption	3.6	Three		3/
Flammability	3.6	Three	X	
Fluid resistance	3.6	3/		3/
Heat shock	3.6	Three	X	
Restricted shrinkage	3.6	Three	X	
Modulus	3.6	Three		X

1/ Nonconformance of one determination shall constitute failure of the sample. A description of the numerical point of the failure shall also be recorded.

2/ All values on which the average is based shall be recorded.

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3/ Sample must pass all requirements in the specification sheet. See tensile strength, ultimate elongation, and dielectric strength (as applicable) for number of determinations per sample and how to report results.

4.6.2 ASTM tests. ASTM tests shall be performed in accordance with ASTM Standard Methods of Test (see 2.2) with any exceptions that may be specified herein or in the specification sheets.

4.6.3 Shrinkage conditions

4.6.3.1 Unrestricted. Six inch lengths of sleeving shall be permitted to recover at standard condition for the time specified in the applicable specification sheet. Specimens shall be examined as specified.

4.6.3.2 Restricted. A length of sleeving shall be placed over the mandrel shown in Figure 1 in a manner that will, after shrinking, slightly expose each end for electrical connections. Shrinkage shall be as specified in 4.6.3.1. After the recovery period, the specimen shall be observed for fit and cracking. Any movement on the mandrel or cracks shall be cause for rejection. The specimen shall then be used for the voltage withstand test.

4.6.4 Dimensions and tolerances. Micrometer measurements shall be made in accordance with ASTM D 374.

4.6.4.1 Inside diameter (I.D.).

4.6.4.1.1 I.D. (as supplied). To measure the I.D. of the sleeving, select a smooth gage rod that has a diameter equal (plus 0, minus 0.002 inch, or 2 percent, whichever is less) to the minimum acceptable I.D. of the "as supplied" sleeving. The I.D. shall be considered acceptable when the gage rod shall pass freely into the sleeving sample without expanding the wall of the sleeving.

4.6.4.1.2 I.D. after unrestricted shrinkage. Sleeving shall be shrunk as specified in 4.6.3.1. Select a smooth gage rod that has a diameter equal (minus 0, plus 0.002 inch, or 2 percent, whichever is less) to the maximum acceptable sleeving I.D. after unrestricted shrinkage. The max I.D. dimension of the sleeving shall be considered acceptable when the wall of the sleeving is expanded by the insertion of the gage rod, when there is no visible air space between the end of the sleeving and the rod, or when the gage rod cannot be inserted in the sleeving.

4.6.4.2 Wall thickness. Wall thickness shall be determined by one of the following methods:

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metal rod. The outside diameter of the metal rod shall be the maximum

4.6.4.2.1 Micrometer method. After the unrestricted shrinkage specified in 4.6.3.1, the sleeving shall be slit lengthwise into two hemicylindrical pieces. A smooth gage rod shall be selected, measured, and placed in contact with the inside surface of the sleeving. The total thickness of the gage rod plus sleeving wall shall then be measured. The maximum and minimum wall thickness shall be calculated by subtracting the gage rod measurement from the measurement of the gage rod plus sleeving wall.

4.6.4.2.2 Optical method. The wall thickness of the sleeving, after the unrestricted shrinkage specified in 4.6.3.1 shall be measured with a contour projector as specified in MIL-STD-120 or with a calibrated microscope.

4.6.4.3 Longitudinal change. The sleeving shall be measured in the "as supplied" condition and after the unrestricted shrinkage specified in 4.6.3.1 Longitudinal change shall be calculated as percent. Measurements shall be made with an accurate rule.

L_1 = original length

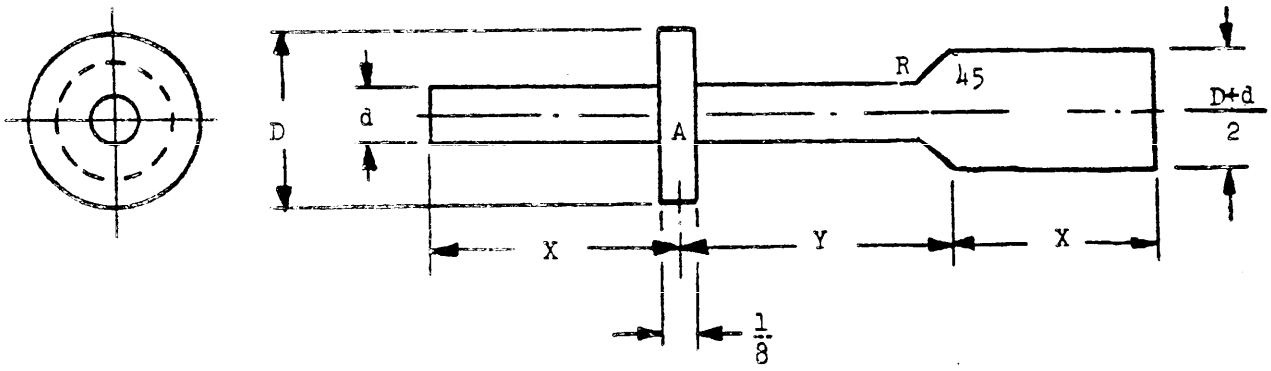
L_2 = length after unrestricted shrinkage

$$\% \text{ change} = \frac{L2 - L1}{L1} \times 100$$

4.6.5 Voltage withstand. Using the specimen prepared in 4.6.3.2, tightly wrap the first layer of the outer electrode, a strip of metal foil, around the sleeving specimen. The outer electrode shall conform closely to the shape of the sleeving particularly at the areas A and R shown in Figure 1. Allow sufficient sleeving area at each end to remain unwrapped so that flashover between the mandrel and the foil will not occur. Wrap two or more layers of foil over the first wrap allowing a free end of 1/2 inch for connection of the electrode. 2000 volts shall be applied for one minute in accordance with the test procedure of ASTM D 149.

4.6.6 Low temperature flexibility. For sleeving sizes less than 0.400 inch (max I.D. nominal after unrestricted shrinkage), the sleeving shall be shrunk and conditioned as specified in 4.6.3.2 onto a stranded AWG (nearest AWG wire) which is larger that the sleeving max I.D. nominal after unrestricted shrinkage). For sleeving sizes 0.400 inch and larger, a 6 by 0.25 inch longitudinal strip shall be cut from sleeving that has been shrunk and conditioned as specified in 4.6.3.1. The specimen (strip or sleeving) and a steel mandrel of the size specified in Table V shall be conditioned in a cold chamber for four hours at the temperature specified in the applicable specification sheet. After completion of the conditioning period and while still in the cold chamber at the specified temperature, the specimen shall be bent around the mandrel through not less than 360 degrees. Unless otherwise specified in the specification sheet, the time to complete the bend shall be 10 $\pm$ 2 seconds. Visually examine the sleeving for cracks.

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d = max. I.D. of sleeving, nominal, after unrestricted shrinkage $^{+5}_{-0}$ percent.

D = min. I.D., nominal, of "as supplied" sleeving $^{+5}_{-0}$ percent.

A = That part of the mandrel shaped as shown with a width of 1/8 inch and a diameter D.

B = That part of the mandrel shaped as shown with a 45° angle between the $\frac{D + d}{2}$ and d dimensions.

All edges involved in this mandrel shall have no burrs.

Section B can be provided with a longitudinal V-groove to facilitate removal of the sample with a knife or razor blade after the testing is complete.

Max I.D. of sleeving (nominal) after unrestricted shrinkage in inches	Mandrel section in inches	
	X	Y
Less than 0.050 1/	0.50	0.25
0.050 to 0.125	0.50	0.25
0.126 to 0.374	1	0.50
0.375 to 2.000	2	2
2.001 to 3.000	3	3

1/ For sleeving sizes less than 0.050 I.D. inch (after unrestricted) shrinkage), a straight cylindrical mandrel shall be made with an O.D. equal to D.

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TABLE V. Mandrel size for bend tests

Nominal sleeving I. D. (max., after unrestricted shrinkage inches)	Mandrel diameter 1/ inches
0.023 to 0.125	5/16 ± 0.002
0.126 to 0.250	3/8 ± 0.003
0.251 to 1.000	7/16 ± 0.004
1.001 to 2.000	7/8 ± 0.005
2.001 to 3.000	1 ± 0.005

1/ Unless otherwise specified in the specification sheet.

4.6.7 Heat shock. A 6 inch length of the sleeving or a 6 by 1/2 inch strip cut longitudinally from sleeving that has a nominal I.D. greater that 1/2 inch as supplied shall be shrunk as specified in 4.6.3.1, then placed in the oven for 4 hours at the temperature specified in the specification sheet. The specimen may be suspended vertically in the oven by means of a wire attached to a hole in the specimen. While in the oven and after removal from the oven, the specimen shall be examined for evidence of cracking, flowing or dripping. The specimen shall then be conditioned as specified in 4.6.1.1 and bent through 360 degrees in two to four seconds over the steel mandrel specified in Table V. The specimen shall then be visually examined for cracks. Disregard longitudinal side cracking.

4.6.8 Heat resistance. The sleeving shall be shrunk and conditioned as specified in 4.6.3.1. A suitable number of four inch lengths of the sleeving shall then be placed in an oven at the temperature and for the time specified in the specification sheet. After removal from the oven, the sleeving shall be conditioned as specified in 4.6.1.1. The sleeving shall then be tested for conformance to the ultimate elongation, tensile strength, and dielectric strength requirements (as applicable) of the specification sheet.

4.6.9 Corrosion. Prior to corrosion testing, the sleeving sample shall be shrunk as specified in 4.6.3.1. The applicable corrosion test procedure(s) specified in the specification sheets shall be followed.

4.6.9 Corrosion in contact with copper. A six inch length of shrunk sleeving shall be slid over a straight, clean, bare copper conductor. For sleeving of 0.25 inch inside diameter and smaller (max. I.D. after unrestricted shrinkage), a single conductor shall be used; for larger sleeving, a twisted bundle of several conductors shall be used, each AWG 18 or smaller. The specimens on the conductor shall be heated in an air-circulating oven for the time and the temperature specified in the applicable specification sheet. The specimen shall then be removed from the oven, cooled, and conditioned in accordance with 4.6.1.1. The sleeving shall be slit open and the copper examined for pitting and blackening. The sleeving shall be considered noncorrosive only if the copper is not pitted or blackened. Darkening of the copper due to normal air oxidation shall not be cause for rejection.

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4.6.9.2 Copper mirror corrosion. For sizes 0.400 inch and larger (max. I.D. nominal after unrestricted shrinkage), a 1/4 by 1 inch strip shall be cut longitudinally from the shrunk sleeving. For sizes smaller than 0.400 inch (max. I.D. nominal after unrestricted), a number of pieces of shrunk sleeving having an area of approximately 0.25 square inch on one side shall be used. The strip or piece of sleeving shall be placed in the bottom of a clean, dry 1/2 by 12 inch test tube. A copper glass mirror about 1/4 inch by 1 inch and a calibrated thermometer shall be suspended 6 to 7 inches above the bottom of the test tube by means of a fine, clean copper wire attached to a cork that has been wrapped in clean aluminum foil. A second clean test tube containing only the mirror and the thermometer shall be used as a control. The lower two inches of the test tube shall be immersed in an oil bath at the temperature and for the time specified in the applicable specification sheet. The temperature of that portion of the test tube containing the copper mirror shall be below 60°C. After removal from the oil bath and cooling to room temperature, visually examine the mirror for corrosion by placing it on a white background in good light. Evidence of corrosion shall be the removal of the copper film from any part of the mirror as evidenced by the white background showing through. Disregard removal of copper from the bottom 1/16 inch of the mirror. Discoloration of the copper film or reduction of its thickness should not be considered corrosion. The mirrors shall be vacuum deposited copper with a thickness equal to 10 ± 5 percent transmission of normal light of 5000 Angstroms. They shall be stored in a vacuum and shall be used only if no oxide film is present and the copper is not visually damaged or contaminated.

4.6.10 Fluid resistance. The sleeving shall be shrunk as specified in 4.6.3.1. A suitable number of 4 inch lengths of sleeving shall be totally immersed in each of the fluids specified in Table VI. The volume of the fluid shall not be less than 20 times that of the sleeving. After 24 hours immersion at 24 ± 3°C (75 ± 5.4°F), the sleeving shall be lightly wiped and then conditioned for 45 ± 15 minutes at room temperature. The sleeving from each of the test fluids shall then be tested for conformance to the tensile strength, ultimate elongation, and dielectric strength (as applicable) of the specification sheet.

Table VI. Test Fluids

Fluid	Conforming to
Hydraulic fluid, petroleum base	MIL-H-5606
JP-4 fuel	MIL-T-5624
Lubricating oil	MIL-L-7808
Lubricating oil	MIL-L-23699
5% NaCl	SS-S-550
Deicing fluid	MIL-A-8243
Cleaning fluid, aircraft	MIL-C-43616

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4.6.11 Odor. The sleeving shall be shrunk (4.6.3.1) and conditioned as specified in 4.6.1.1. Place a quantity of shrunk sleeving that has an approximate area of 10 square inches (measured on one side) in an airtight steel container. After 24 hours at room temperature, open the container, and carefully smell any odor that may be present. An objectionable pungent odor shall be cause for rejection.

4.6.12 Modulus

4.6.12.1 Secant modulus. Full sections of sleeving shall be tested when the min. I.D. as supplied is 0.25 inch or less. For larger sizes, a 0.25 inch strip shall be cut longitudinally from the sleeving. No metal plugs are necessary when testing full sections of the sleeving. The testing procedure and percent strain shall be as specified in the specification sheet.

4.6.12.2 Tensile stress. Full sections of sleeving shall be tested when the min. I.D. of the sleeving to be tested is 0.25 inch or less. For larger sizes a 0.25 inch strip shall be cut longitudinally from the sleeving No metal plugs are necessary when testing full sections of sleeving. The testing procedure and percent strain shall be as specified in the applicable specification sheet. One inch bench marks and 2 inch initial jaw separation shall be used. A break at a bench mark or outside the gage length shall be cause for retest.

4.6.13 Tensile strength and Ultimate elongation. The sleeving shall be shrunk as specified in 4.6.3.1. Full sections of sleeving shall be tested when the max. I.D., after unrestricted shrinkage, is 0.25 inch or less. For larger sizes, cut a 0.25 inch strip longitudinally from the sleeving. No metal plugs are necessary when testing full sections of sleeving. The test procedure for tensile strength and ultimate elongation shall be as specified in the specification sheet. One inch bench marks and 2 inch initial jaw separation shall be used. Testing speed shall be as specified in the specification sheet. A break at the bench mark or outside the gage length shall be cause for retest.

4.6.14 Flammability. One of the following shall be performed as specified in the specification sheet. Sleeving that passes the test specified in 4.6.14.3 has better flame resistant properties than sleeving that passes either 4.6.14.1 or 4.6.14.2 and sleeving that passes 4.6.14.2 has better flame resistant properties than sleeving that passes 4.6.14.1.

4.6.14.1 An 18 inch specimen of sleeving shall be shrunk as specified in 4.6.3.1 onto a 21 inch metal rod. The outside diameter of the metal rod shall be the maximum recovered inside diameter of the sleeving specified in the applicable specification sheet. The specimen so prepared shall be tested in accordance with FED-STD-228 Method 5221 for Vertical Flamability. Brown scorching or soot on the indicator flag shall not be cause for rejection.

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4.6.14.2 A 22 inch specimen of sleeving shall be shrunk as specified in 4.6.3.1 and tested in accordance with ASTM D 876.

4.6.14.3 A 26 inch length of sleeving shall be shrunk on a .015 to .032 spring steel music wire. The specimen shall then be mounted in the same manner and test equipment described in 4.6.14.1. The sleeving shall be mounted by clamping the top so there shall be no chimney effect. There shall also be a flat horizontal layer of untreated surgical cotton placed on the floor of the enclosure (centered under the specimen 9-1/2 inches below the point of flame application). The specimen so prepared shall then be tested in accordance with FED-STD-228 Method 5221 for Vertical Flammability, with the following exceptions:

The valve supplying gas to the burner is to be opened to apply the flame to the specimen automatically. This valve is to be held, open for 15 seconds, closed for as little as 15 seconds, opened for 15 seconds, etc. for a total of five 15 second applications of the gas flame to the specimen with as little as 15 seconds between applications. In no case is the gas flame to be reapplied to the specimen until any flaming or glowing of the specimen from the previous application of the gas flame ceases of its own accord even though waiting for the specimen to cease flaming or glowing may result in more that 15 second between applications of the gas flame. The duration of flaming or glowing of the specimen after each removal of the gas flame is to be recorded.

Note is to be taken and recorded of whether any flaming or glowing particles or flaming drops fall from the specimen.

If more than 25 percent of the indicator flag is burned away or charred (brown scorching and soot are to be ignore) after all of the five applications of flame or if flaming or glowing particles or flaming drops fall from the specimen at any time and ignite the cotton (flameless charring of the cotton is to be ignored), the specimen shall be judged to have conveyed flame. Any tubing from which a specimen continues to flame or glow longer than one minute is not acceptable.

4.6.15 Color stability. A 4 inch specimen of sleeving, shrunk as specified in 4.6.3.1, shall be placed in an air-circulating oven for the time and at the temperature specified in the applicable specification sheet. After removal from the oven, the specimens shall be cooled, conditioned in accordance with 4.6.1.1, and examined for conformance to the color requirement of the specification sheet.

4.6.16 Shelf Life. The sleeving in original unopened containers shall be stored for a period of one year, at temperatures of 18 to 35°C (65 to 95°F). After the storage period, the sleeving shall be examined as specified in 4.6.4 for conformance to 3.7.

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5. PACKAGING

5.1 Preservation - packaging. Preservation packaging shall be level A or C as specified (See 6.2.1).

5.1.1. Level A

5.1.1.1 Unit package. Nonheat shrink sleeving, cut to predetermined lengths, furnished to this specification shall be packaged in cans conforming to Type IV, class 2 of PPP-C-96, or in heat sealable pouches conforming to Type III, Class E, style 1 of MIL-B-117. Plastic lids shall be furnished for resealing the cans. Quantity per package shall be as specified (See 6.2.1).

5.1.2 Level C. The sleeving shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (See 6.2.1).

5.2.1 Level A. Nonheat shrink sleeving packaged as specified in 5.1 shall be packed in containers conforming to style RSC, V2s or V3s of PPP-B-636; Class 3 of PPP-B-585; Class 2 of PPP-B-621; or PPP-B-601 overseas type. Cans shall be packed in the quantity and arranged as specified in PPP-C-96. Boxes shall be closed and reinforced in accordance with the appendix to the applicable container specification.

5.2.2 Level B. Nonheat shrink sleeving packaged as specified in 5.1 shall be packed in fiberboard or woodcleated fiberboard boxes conforming to Type CF or SF, class domestic, style RCS of PPP-B-636; class 1 or 2 of PPP-B-585, class 1 of PPP-B-621; domestic type of PPP-B-601; or class 1 of PPP-B-591. Cans shall be packed in the quantity and arrangement specified in PPP-C-96. Boxes shall be closed in accordance with the appendix to the applicable container specification.

5.2.3 Level C. Interior packages shall be packed in containers of the size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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3.3.1 Additional markings.

a. Date manufactured
b. Expiration date

6. NOTES

6.1 Intended use. Nonheat shrink sleeving is intended for use as a snug fitting electrical insulator over wire bundles, splices, cables, cable harnesses, bus bars, connectors, and terminals, metal or fibrous tubing, hydraulic line couplers, high voltage tubes, flexible wave guides and rectifier leads, as extra insulation over hot spot areas, and as cable blast shield in rocket launching This sleeving should be considered for use as a substitute for heat shrinkable sleeving when it has been determined that the application of heat for shrinking purposes could result in damage to a heat sensitive part or area of assembly.

6.1.1 Non-heat shrink sleevings are chemically expanded through the incorporation of a solvent into the sleeving. Upon exposure to the atmosphere, the solvent evaporates, thereby causing the sleeving to recover (shrink) to its pre-determined size. Because of the evaporation rates, it is recommended that individual packaging contain only that amount required for immediate use. For volume applications, use material applied in resealable containers.

NOTE

The solvents employed in the fabrication of nonheat shrinkable sleeving are flammable. Care should be exercised in their use through total recovery. Upon completion of the shrinkdown period (total evaporation of solvent) the sleeving is required to meet the flammability requirements contained herein.

6.2 Ordering Data

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet.

b. Part number (including color designation) (See specification sheet and 1.2.2)

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c. Quantity, total footage (unit of issue = 1 foot)

d. Precut sleeving lengths desired.

e. Number of lengths per package.

f. Packing and Packaging requirements.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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